Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The straight chain polymer is formed by:

(A) hydrolysis of CH_3SiCl_3 followed by condensation polymerisation

(B) hydrolysis of $(CH_3)_4$ Si by addition polymerisation

(C) hydrolysis of $(CH_3)_2 SiCl_2$ followed by condensation polymerisation

(D) hydrolysis of $(CH_3)_3 SiCl$ followed by condensation polymerisation

Q.2

An example of a double salt is

(A) Bleaching powder

(B) $K_4 [Fe(CN)_6]$

(C) Hypo

(D) Potash Alum

Q.3

The substance used as a smoke screen in warfare is

(A) $SiCl_4$

(B) PH_3

(C) PCl_5

(D) Acetylene

Q.4

Which of the following facts regarding boron and silicon is not true?

(A) Boron and silicon are semiconductors

(B) Boron and silicon form halides which are not hydrolysed

(C) Boron and silicon react with magnesium to form magnesium boride and magnesium silicide which are decomposed by acids to give volatile borane and silane, respectively

(D) Both boron and silicon react with alkalis to form borates and silicates containing BO_4 and SiO_4 tetrahedral units, respectively

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Q.5

Which of the following statements about H_3BO_3 is not correct ?

(A) It is a strong tribasic acid

(B) It is prepared by acidifying an aqueous solution of boraxH_3BO_3 is a weak monobasic acid.

(C) It has a layer structure in which planar BO_3 units are joined by hydrogen bonds

(D) It does not act as proton donor but acts as a Lewis acid by accepting a lone pair of electrons

Q.6

In graphite, electrons are

(A) Localised on every third C-atom

(B) Present in anti-bonding orbital

(C) Localised on each C-atom

(D) Spread out between the structure

Q.7

Glass reacts with HF to produce

(A) SiF_6

(B) SiF_4 (Silicon tetra fluoride).

(C) H_2SiO_3

(D) Na_3AlF_6

Q.8

Which of the following equations is not correctly formulated?

(A) $Na_2\left[B_4O_5(OH)_4\right]\cdot 8H_2O + 2HCl \rightarrow 2NaCl + 4H_3BO_3 + 5H_2O$
(borax)

(B) $2BN + 6H_2O \rightarrow 2H_3BO_3 + 2NH_3$

(C) $H_3BO_3 \xrightarrow[-H_2O]{375\ K} HBO_2$; $4HBO_2 \xrightarrow[-H_2O]{435\ K} H_2B_4O_7$; $H_2B_4O_7 \xrightarrow[-H_2O]{\text{red heat}} 2B_2O_3$
(metaboric acid) (tetraboric acid) (Boric oxide)

(D) H_3BO_3 is a weak monobasic acid as it liberates hydrogen ions as
$H_3BO_3 \rightarrow H^+ + H_2BO_3^-$

Q.9

Al_2O_3 can be converted to anhydrous $AlCl_3$ by heating

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(A) Al_2O_3 with $NaCl$ in solid state

(B) a mixture of Al_2O_3 and carbon in dry Cl_2 gas

(C) Al_2O_3 with Cl_2 gas

(D) Al_2O_3 with HCl gas

Q.10

The tendency of BF_3, BCl_3 and BBr_3 to behave as Lewis acid decreases in the sequence

(A) $BCl_3 > BF_3 > BBr_3$

(B) $BBr_3 > BCl_3 > BF_3$

(C) $BBr_3 > BF_3 > BCl_3$

(D) $BF_3 > BCl_3 > BBr_3$

Q.11

Name the type of the structure of silicate in which one oxygen atom of $[SiO_4]^{4-}$ is shared ?

(A) Linear chain silicate

(B) Sheet silicate

(C) Pyrosilicate

(D) Three dimensional

Q.12

Which of the following structure is similar to graphite?

(A) B

(B) B_4C

(C) B_2H_6

(D) BN

Q.13

The precipitate of $Al(OH)_3$ dissolves in $NaOH$ solution. It is due to the formation of

(A) $Al\left(H_2O\right)_4\left(OH\right)_2^+$

(B) $Al\left(H_2O\right)_3\left(OH\right)_3$

(C) $Al\left(H_2O\right)_2\left(OH\right)_4^-$

(D) $Al(OH)_6^{3-}$

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Q.14

Heating of diborane with excess of ammonia to a high temperature yields

(A) borane

(B) diammoniate of borane

(C) borazine

(D) borone nitride

Q.15

Which of the following is known as butter of tin?

(A) $SnCl_2 \cdot 2H_2O$

(B) $SnCl_2 \cdot 3H_2O$

(C) $SnCl_2 \cdot 4H_2O$

(D) $SnCl_2 \cdot 5H_2O$

Q.16

Which of the following equations is not actually observed?

(A) $4Sn + 10\underset{\text{(dilute)}}{HNO_3} \rightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$

(B) $3Sn + 8\underset{\text{(dilute)}}{HNO_3} \rightarrow 3Sn(NO_3)_2 + 2NO + 4H_2O$

(C) $Sn + \underset{(\text{conc.})}{2HCl} \rightarrow SnCl_2 + H_2$

(D) $Sn + 2\ \underset{\text{(conc.)}}{2H_2\ SO_4} \rightarrow SnSO_4 + 2H_2O + SO_2$

Q.17

Which of the following fuel gases has the highest calorific value?

(A) Water gas

(B) Coal gas

(C) Producer gas

(D) Semi-water gas

Q.18

Self protective metal among the following is:

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(A) Na

(B) Al

(C) Fe

(D) K

Q.19

Boron halides are ____

(A) Electron deficient compounds

(B) Ionic compounds

(C) Lewis bases

(D) Used as refractory compounds

Q.20

Which of the following statements is not correct?

(A) Lead(II) chloride and lead(II) iodide are soluble in hot water but crystallize out on cooling

(B) Lead(IV) chloride cannot be separated from the solution as it readily decomposes to lead(II) chloride and chlorine

(C) Lead(IV) chloride is not readily hydrolysed by water

(D) Per cent of lead(II) halides decreases with the increase in atomic number of the halogen

Q.21

In aluminate ion, the coordination number of aluminium is:

Q.22

Average atomic weight of boron is 10.10 and boron exists in two isotopic forms B^{10} and B^{11}. The percentage abundance of B^{10} is?

Q.23

In diborane, the sum of two H-B-H angles are nearly :

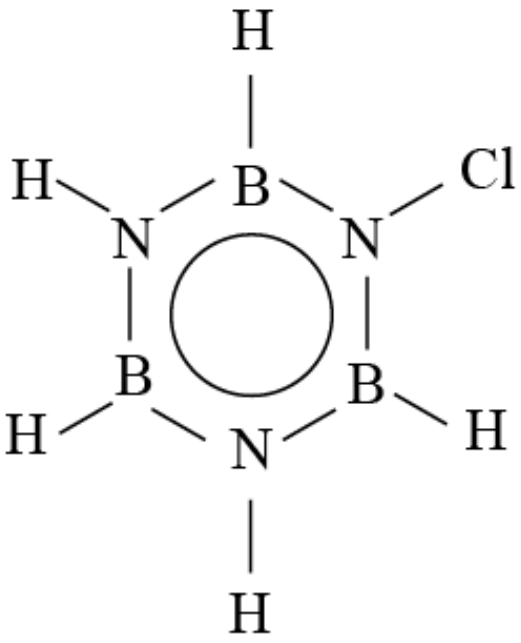
Q.24

The number of water of crystallization in borax is

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Q.25

How many derivate structures of $B_3\ N_3H_4X_2$ can be derived from the basic structure by the replacement of two hydrogen atoms?



Q.26

What is the change in oxidation number of carbon in the following reaction?

$$CH_4(g) + 4Cl_2(g) \longrightarrow CCl_4(l) + 4HCl(g)$$

Q.27

The number of oxides formed by carbon is:

Q.28

The charge $-x$ of $Si_6O_{18}^{x}$ will be

Q.29

Aluminium chloride exists as dimer, Al_2O_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives how many number of ions?

Q.30

The oxidation state of carbon in ionic carbide is x. The value of $x + 10$ is:

Answers & Solutions

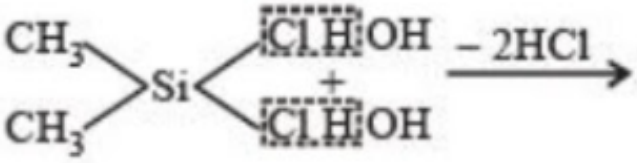
Q.1 Answer:

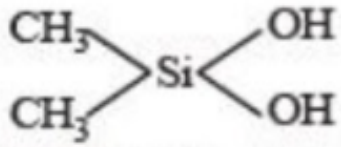
hydrolysis of $(CH_3)_2\ SiCl_2$ followed by condensation polymerisation

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Solution:

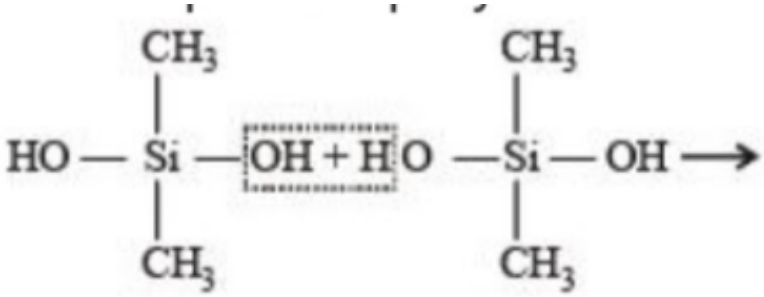
Hydrolysis of substituted chlorosilanes yield corresponding silanols which undergo polymerisation.

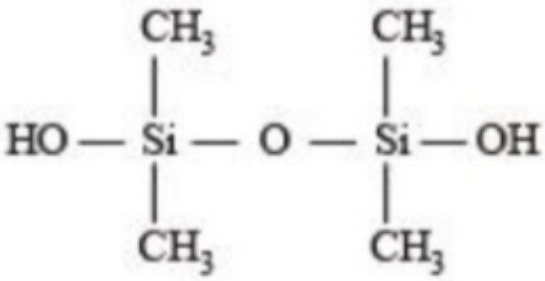




Dialkyl silandiol

Polymerisation of dialkyl silandiol yields linear thermoplastic polymer.





Q.2 Answer:

Potash Alum

Solution:

Potash Alum, $K_2SO_4.Al_2(SO_4).24H_2O$ **is a double salt.**

Q.3 Answer:

$SiCl_4$

Solution:

$SiCl_4$ **gets hydrolysed in moist air and gives white fumes which are used as a smoke screen in warfare.**

Q.4 Answer:

Boron and silicon form halides which are not hydrolysed

Solution:

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The halides of B and Al are hydrolysed.

$$BCl_3 + 3H_2O \rightarrow \underset{\text{Boric acid}}{B(OH)_3} + 3HCl \text{ and } SiCl_4 + 4H_2O \rightarrow \underset{\text{Silicic acid}}{Si(OH)_4} + 4HCl$$

Q.5 Answer:

It is a strong tribasic acid

Solution:

1. It is not a tribasic acid.

Option (a) is incorrect for H_3BO_3

2. Aqueous solution, instead of donating the $-OH$ group it breaks the $H-OH$ bond of water and accepts the $-OH$ group from it leaving in the solution hence it is mono basic acid.

Q.6 Answer:

Spread out between the structure

Solution:

In graphite, each carbon is sp^2-hybridized and the single occupied unhybridized porbitals of C-atoms overlap side wise to give π-electron cloud which is delocalized and thus the electrons are spread out between the structure.

Q.7 Answer:

SiF_4 (Silicon tetra fluoride).

Solution:

Common glass consists of silica (SiO_2). When HF reacts with silica it will form H_2SiF_6 or SiF $_4$ Both can soluble in water which means that a fresh surface of SiO_2 will be exposed to attack by HF.

Q.8 Answer:

H_3BO_3 is a weak monobasic acid as it liberates hydrogen ions as

$H_3BO_3 \rightarrow H^+ + H_2BO_3^-$

Solution:

H_3BO_3 is a weak monobasic acid. It does not liberate H^+ but accepts OH^-, i.e. it is Lewis acid.

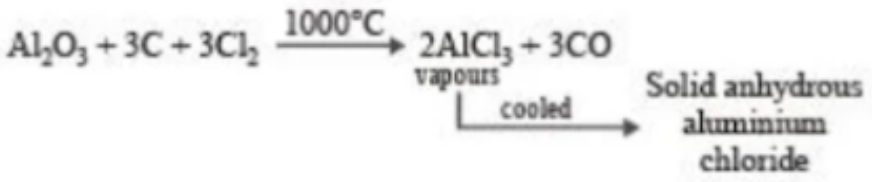
Q.9 Answer:

a mixture of Al_2O_3 and carbon in dry Cl_2 gas

Solution:

Al_2O_3 can be converted to anhydrous $AlCl_3$ by heating a mixture of Al_2O_3 and carbon in dry Cl_2

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Q.10 Answer:

$BBr_3 > BCl_3 > BF_3$

Solution:

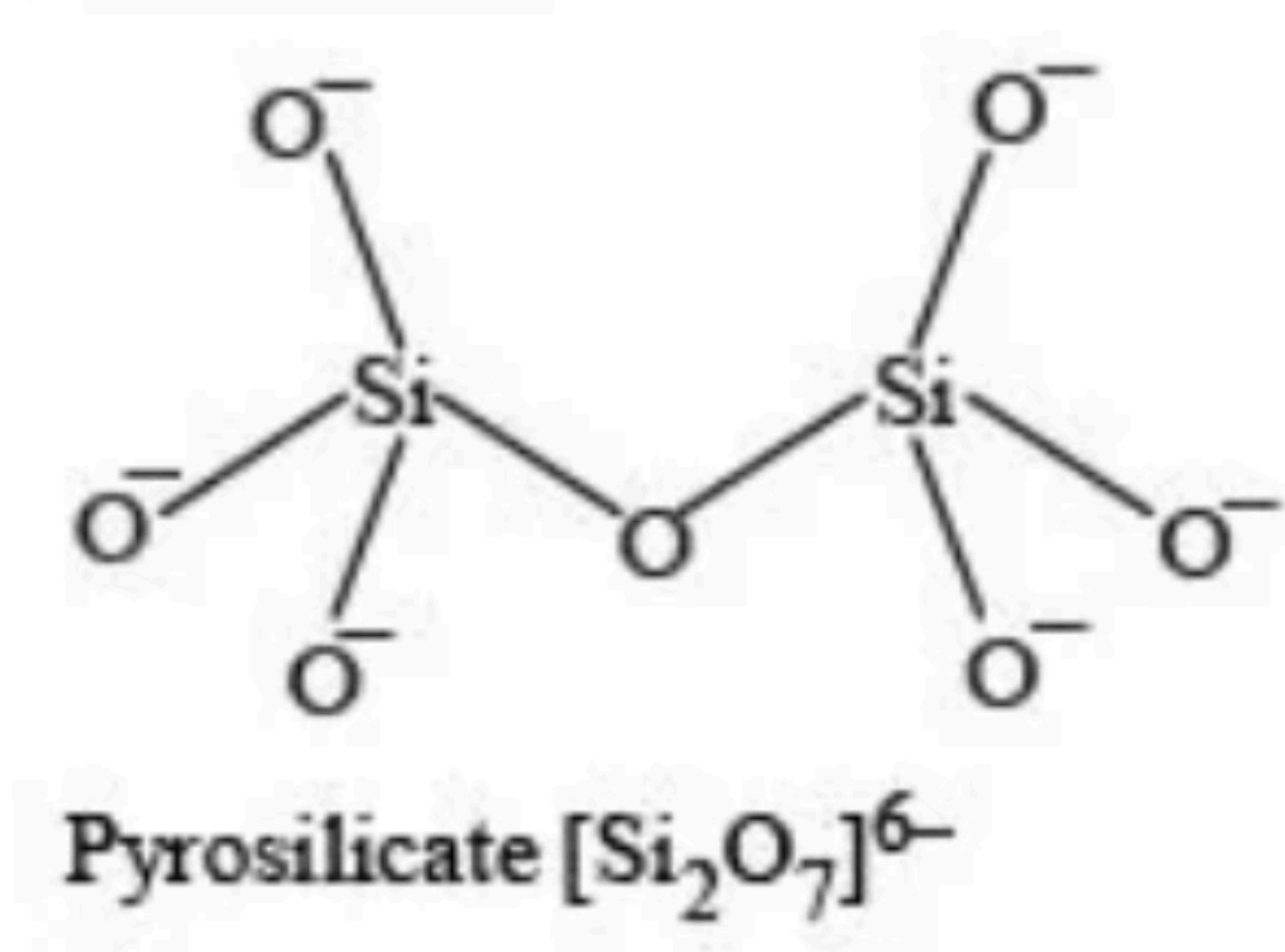
A molecule is said to be a lewis acid if it can accept a lone pair.

When the halide is more electronegative the interelectronic repulsion increases making the molecule difficult to accept a lone pair. Size of halide also behaves in the same way. Also the degree of hydrolysis increases from fluorine to iodine,so the lewis acidic character increases in the same order, So,the order of lewis acidity is $BBr_3 > BCl_3 > BF_3$

Q.11 Answer:

Pyrosilicate

Solution:



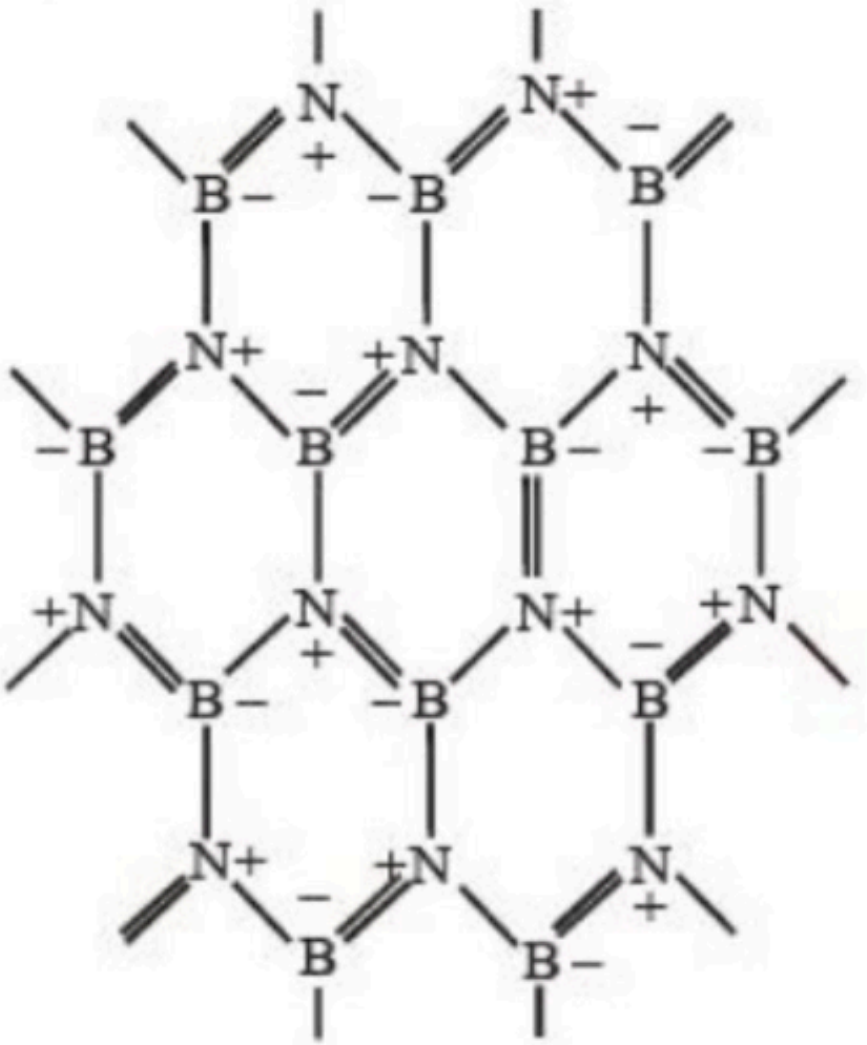
Q.12 Answer:

BN

Solution:

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Boron nitride (BN) is known as inorganic graphite. The most stable form is hexagonal one. It has layered structure similar to graphite.



Q.13 Answer:

$\mathrm{Al}\left(\mathrm{H_2O}\right)_2\left(\mathrm{OH}\right)_4^-$

Solution:

Dissolution of $\mathrm{Al(OH)_3}$ by a solution of NaOH produces a complex compound shown as below:

$\mathrm{Al(OH)_3 + NaOH \longrightarrow \left[Al\left(H_2O\right)_2\left(OH\right)_4\right]^-}$

Q.14 Answer:

borazine

Solution:

When diborane is heated with excess ammonia, it reacts with ammonia to form borazine and hydrogen.

"This reaction takes place at a high temperature of $180-190^{\circ}\mathrm{C}''$. This reaction of diborane and ammonia is considered to be highly complex in nature. Diborane is majorly used as rocket propellants and also as a reducing agent.

The chemical equation for the reaction is:

$\mathrm{B_2H_6 + NH_3 \rightarrow B_3N_3H_6 + H_2}$

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The balanced equation for the reaction is:

$3B_2H_6 + 6NH_3 \rightarrow 2B_3N_3H_6 + 12H_2$

Q.15 Answer:

$SnCl_2 \cdot 5H_2O$

Solution:

Stannic chloride $(SnCl4)$ is hygroscopic and forms crystalline hydrates containing $3, 5, 6,$ and 8 molecules of water of crystallization. The pentahydrate $SnCl4.5H2O$, is known as "butter of tin" or "oxymuriate of tin." It is used as a mordant for dyes.

Q.16 Answer:

$3Sn + \underset{\text{(dilute)}}{8HNO_3} \rightarrow 3Sn\left(NO_3\right)_2 + 2NO + 4H_2O$

Solution:

\begin{array}{c}
4 \mathrm{Sn}+10 \mathrm{HNO}_{3} \rightarrow 4
\mathrm{Sn}\left(\mathrm{NO}_{3}\right)_{2}+\mathrm{NH}_{4} \mathrm{NO}_{3}+3
**\mathrm{H}_{2} \mathrm{O} **
\text { (dilute) }
\end{array}

Q.17 Answer:

Water gas

Solution:

Water gas is a mixture of CO and H_2, both present in equal amounts. It has the highest calorific value, because both CO and H_2 burn and evolve heat.

Q.18 Answer:

Al

Solution:

Al when comes in contact with air to form Al_2O_3 which forms a protective passive layer on the metal. This layer cannot be removed by chemical means.

Q.19 Answer:

Electron deficient compounds

Solution:

Boron halides are of the form BX_3.

The number of electrons in valence shell of boron is only six (2 short of octet).

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Hence, they are electron deficient compounds.

Q.20 Answer:

Lead(IV) chloride is not readily hydrolysed by water

Solution:

Lead(IV) chloride gets rapidly hydrolyzed by water. $\mathrm{PbCl_4 + 2H_2O \rightarrow PbO_2 + 4HCl}$

Q.21 Answer:

—

Solution:

Aluminate ion isolated is $\mathrm{AlO_2^-}$**.**
When in solution, this is surrounded by 4 water molecules, two Water molecules react with the oxygen to give overall $\mathrm{4OH^-}$ **groups. These four hydroxyl groups and two water molecules are held by coordinate bonds in the complex. Hence the compound has 6 coordinate bonds.**

Q.22 Answer:

—

Solution:

Let the percentage in the form of B^{10} **is** $x\%$ **Therefore, the percentage in the form of** B^{11} **is** $(100-x)\%$ **Average atomic mass** $= x \times \frac{10}{100} + (100-x) \times \frac{11}{100}$

$10.10 = \frac{10 \times x}{100} + \frac{1100-11x}{100}$

$1010 = 1100 - x$

$x = 90$

Therefore, the percentage in which B^{10} **exists is** 90%

Q.23 Answer:

—

Solution:

In diborane, the bonds between boron and the terminal hydrogen atoms are conventional 2-center, 2 -electron covalent bonds. The two $H-B-H$ **angles are nearly** 90° **and** 120°**.**

Q.24 Answer:

—

Solution:

Borax exists in the form of $\mathrm{Na_2B_4O_5(OH)_4 \cdot 8H_2O}$**. thus it has** 8 **water of crystallisation.**

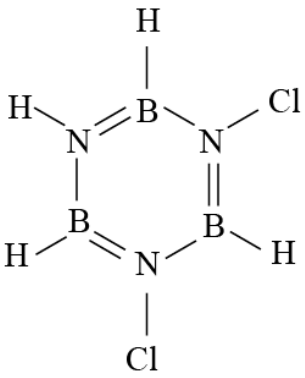
Q.25 Answer:

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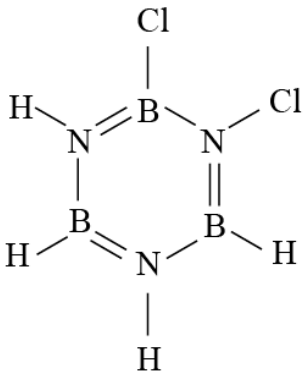
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Solution:

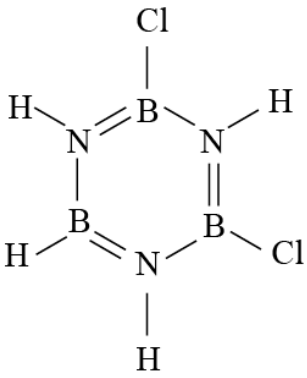
Hence four structures are possible.



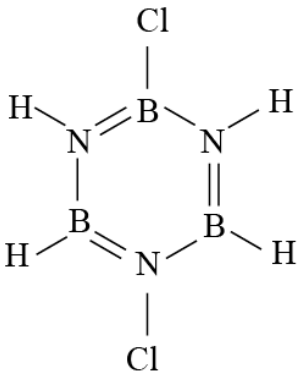
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2



3



4

Q.26 Answer:

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Solution:

$$\overset{-4}{CH_4(g)} + 4\,Cl_2(g) \rightarrow \overset{+4}{CCl_4(l)} + 4HCl(g)$$

The oxidation state of H and Cl are $+1$ and -1 respectively. Therefore the oxidation state of C in methane is -4 and that in carbon tetrachloride is $+4$.

Q.27 Answer:

—

Solution:

Carbon can form three oxides - carbon-mono oxide, carbon dioxide and carbon sub-oxide.

Q.28 Answer:

—

Solution:

$Si_6O_{18}^{-x}$
$= 6 \times 4 - 2 \times 18 = -x$
$= 24 - 36 = -x$
$x = 12$

Q.29 Answer:

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Solution:

$AlCl_3$ **is hydrated in water after dissociation into ions as it behaves as an ionic compound. The interaction can be shown as:**

$$AlCl_3 + 6H_2O \rightarrow [Al\,(H_2O)_6]3 + +3Cl^-$$

It gives 4 **ions**

Q.30 Answer:

—

Solution:

Be_2C **and** Al_4C_3 **are ionic carbides; oxidation state of carbon** $=$

-4

$x = -4$

$x + 10 = (-4) + 10$

$x = 6$

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